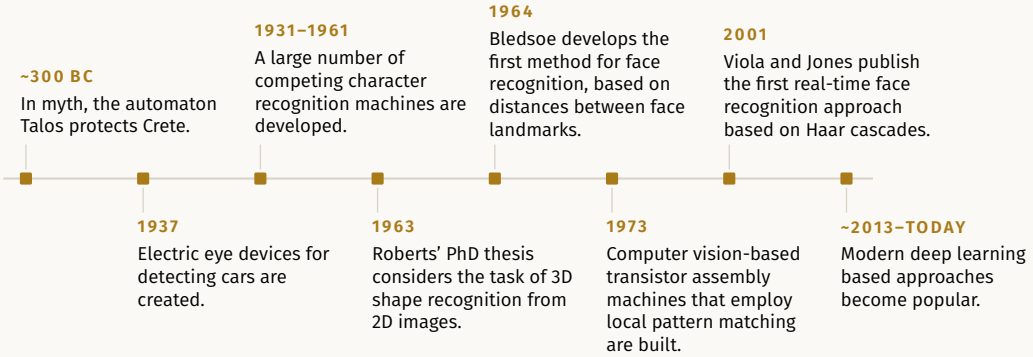
FROM E6691 LECTURE NOTES

History of Deep Learning

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From myth to deep learning in eight milestones



Milestones are evenly spaced, not drawn to scale.

Talos, the automaton of Greek myth

~300 BC

In myth, the automaton Talos protects Crete.

Milestone 1 of 8

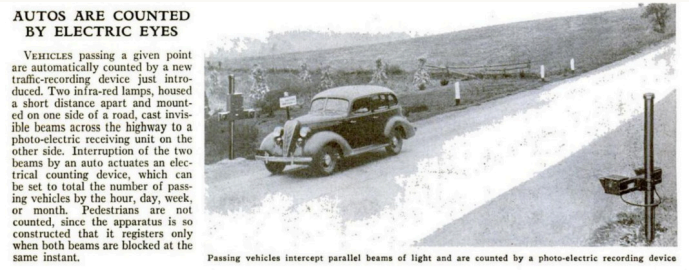


Electric eyes count cars

1937

Electric eye devices for detecting cars are created.

Milestone 2 of 8



AUTOS ARE COUNTED BY ELECTRIC EYES

VEHICLES passing a given point are automatically counted by a new traffic-recording device just introduced. Two infra-red lamps, housed a short distance apart and mounted on one side of a road, cast invisible beams across the highway to a photo-electric receiving unit on the other side. Interruption of the two beams by an auto actuates an electrical counting device, which can be set to total the number of passing vehicles by the hour, day, week, or month. Pedestrians are not counted, since the apparatus is so constructed that it registers only when both beams are blocked at the same instant.

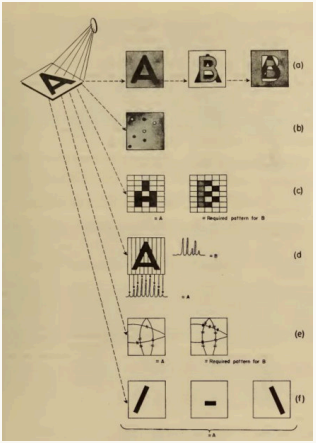
Passing vehicles intercept parallel beams of light and are counted by a photo-electric recording device

Competing character recognition machines

1931–1961

A large number of competing character recognition machines are developed.

Milestone 3 of 8

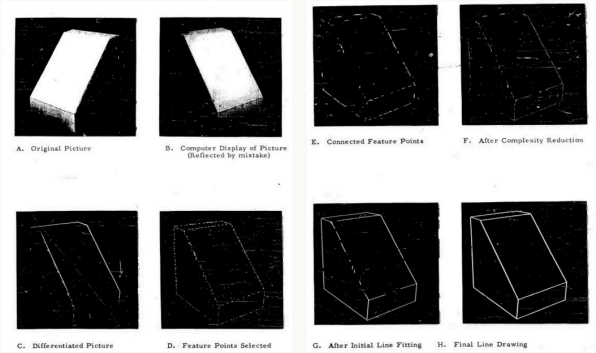


Roberts: 3D shapes from 2D images

1963

Roberts' PhD thesis considers the task of 3D shape recognition from 2D images.

Milestone 4 of 8

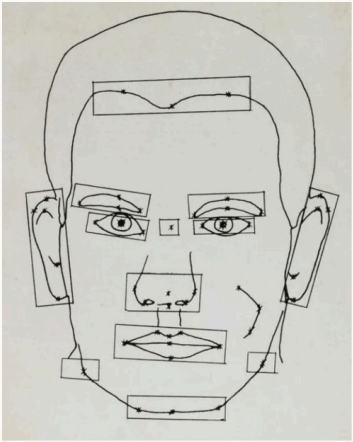


L. G. Roberts, *Machine Perception of Three-Dimensional Solids*, Ph.D. thesis, MIT, 1963. Panels A to H: original picture to final line drawing.

Bledsoe: face recognition from landmark distances

1964

Bledsoe develops the first method for face recognition, based on distances between face landmarks.



Milestone 5 of 8

Vision-guided transistor assembly

1973

Computer vision-based transistor assembly machines that employ local pattern matching are built.

Milestone 6 of 8

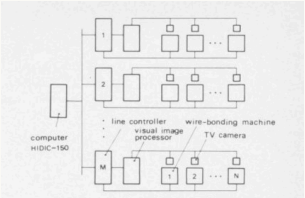


Fig. 2. Automatic transistor assembly system.

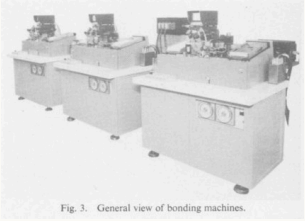


Fig. 3. General view of bonding machines.

Viola and Jones: real-time faces with Haar cascades

2001

Viola and Jones publish the first real-time face recognition approach based on Haar cascades.



P. Viola and M. Jones, *Rapid Object Detection Using a Boosted Cascade of Simple Features*, CVPR 2001.

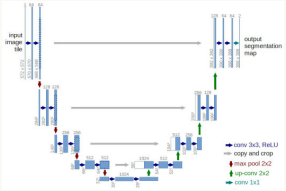
Milestone 7 of 8

Deep learning takes over

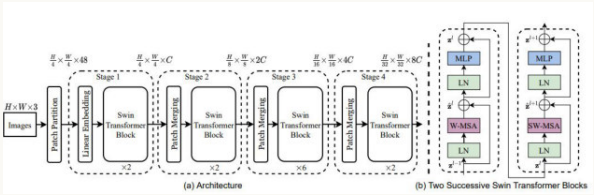
~2013–TODAY

Modern deep learning based approaches become popular.

Milestone 8 of 8



O. Ronneberger, P. Fischer, and T. Brox, *U-Net*, MICCAI 2015.



Z. Liu et al., *Swin Transformer*, ICCV 2021.

Deep convnets broke the plateau on PASCAL VOC

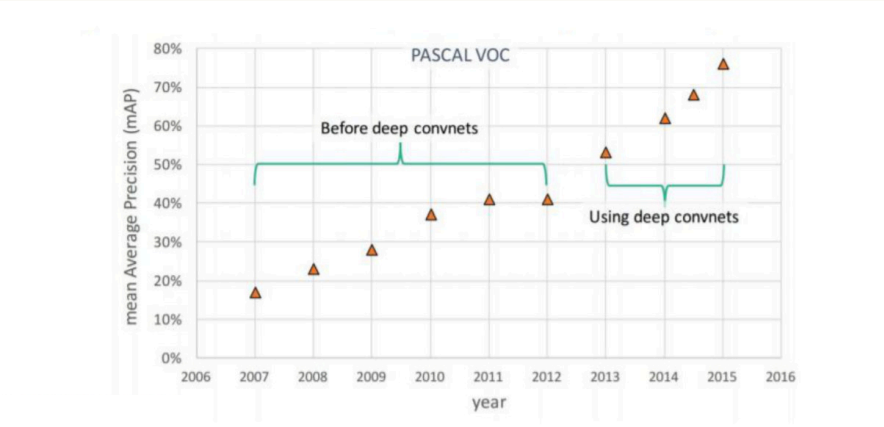
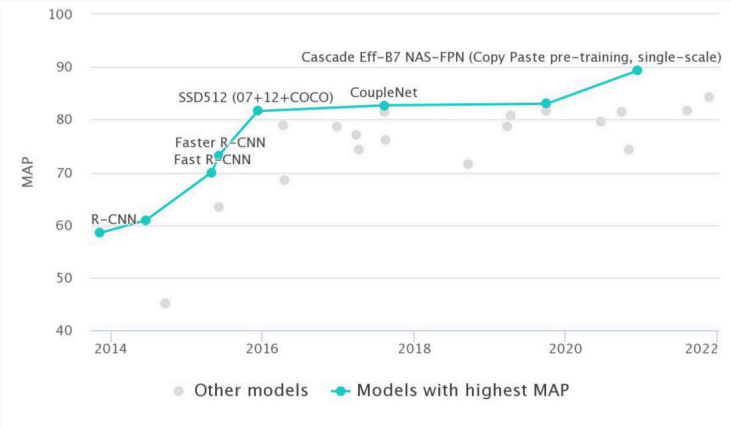


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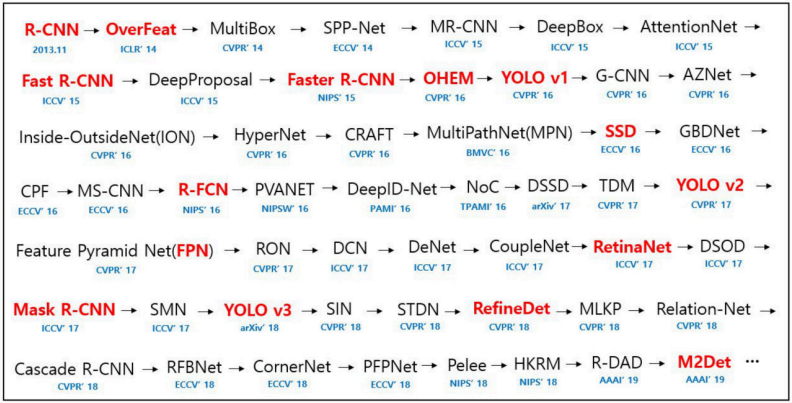
Best reported mAP over time



Datasets

Name	Type	Categories	Image size	Images per category	Objects per image
PASCAL VOC (2007, 2012)	Common objects	20	470×380	303–4087	2.4
MS COCO	Common objects	91	640×480	–	7.3
ImageNet (ILSVRC)	Common objects	21	500×400	–	1.5
KITTI	Driving scene	8	1242×375	–*	–*
Cityscapes	Driving scene	8	2048×1024	–*	–*
...	...	...			...

Snapshot of object detection methods, through 2019



https://github.com/hoya012/deep_learning_object_detection